



ILLUSTRATION BY ANIRBAN GHOSH

integrating large amount of wind and solar power. Continuous expansion and strengthening of the grid is crucial to ensure sufficient transmission capacity is available. The grid must be interconnected to transfer excess generation to states less endowed with renewable energy resources.

All the grid infra must be complemented by demand side flexibility and virtual power plants (VPPs). Demand side flexibility would require regulatory interventions such as time-of-day pricing and smart metering. VPPs can also be promoted by regulators for participation in capacity, ancillary services markets, etc.

A wide array of green technologies would be required for India's low-carbon growth and most of these technologies are emerging, some just confined to laboratories. Therefore, how do we get the promis-

ing technologies to scale up rapidly within a short window? The government should set up cutting edge R&D centres in collaboration with renewable energy companies, businesses, academia and investors. Such innovation hubs could be focussed on advanced solar PV and wind turbine technologies, solar materials, Li+ batteries, alternate energy storage solutions, recycling technologies, smart metering, grid solutions, electrolyzers, green H₂ storage and transportation solutions.

Attracting long-term capital for development of such technologies also depends on stable and effective policies providing financial and demand side support. Production-linked incentive (PLI) schemes for capital-intensive sectors such as solar cells/modules, advanced cell chemistry batteries, EVs, electrolyzers and

fuel cells are worth continuing till we achieve giga scale production to fulfil our growing clean energy needs.

Viability gap funding coupled with demand side assurance for adoption of green H₂ in sectors such as fertilisers, steel, etc., will go a long way in helping it become a mainstream fuel, and thereby eliminate the need for energy from fossil fuels.

Structural challenges associated with discoms' health and land must also be addressed as they threaten to derail the sector. Implementation of the proposed Electricity Act is crucial for the revival of discoms. Direct benefit transfer, discom privatisation, strict RPO compliances and safeguards against contract renegotiation are urgently needed to hasten renewable capacity addition.

The government must undertake various measures for tackling land issues. They can support IPPs by compiling the data of land parcels with high wind and solar generation potential, provide single-window clearances through online applications and undertake digitisation of land records with online maps.

Switching to low-carbon energy is crucial not just to prevent climate change but also because it makes economic sense. It is difficult to say which clean energy will predominate a few decades from now. US scientists achieved a breakthrough in nuclear fusion just a few days ago, creating a vision of a world powered by unlimited, cheap and non-polluting energy.

Nevertheless, energy transition has presented India with a once-in-a-century opportunity to leverage its vast scale and size and channelise its innovative spirit, entrepreneurial and young workforce, and credible institutions into making India a green economy superpower. **BT**

The writer is Chairman and MD of Hero Future Energies